J. Pierce Avner

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EDUCATION:

University of Colorado at Boulder, College of Engineering and Applied Science May 2025
B.S. Computer Science, Minor in Electrical and Computer Engineering
Certified SOLIDWORKS Professional, Certified SOLIDWORKS Additive Manufacturer

- Dean's list (2021-present) **GPA 3.55**
- CU Boulder Esteemed Scholar | Joseph A. Sewall
- Vice Regent | Theta Tau Co-Ed Professional Engineering Fraternity (2023-present)
- Relevant Coursework: Data Science | Software Development | Algorithms | Computer Systems | Principles of Programming Languages | Data Structures | Database Systems | Robotics | Digital Logic

WORK EXPERIENCE:

Tensentric, Boulder, CO **Summer** 2023
Software Engineering Intern

- Developed multithreaded software in c and c++ on Zephyr real time operating system to create autonomous control systems for people with limited dexterity
- Worked with an interdisciplinary team to create and test new algorithms for medical device software under ISO 13485
- Participated in client facing meetings to generate technical and system requirements
- Lead design transfer and testing meetings with clients allowing for a smooth transfer of information and identify improvements for future prototypes

Motion Dance Center, Highlands Ranch, CO 2020-2021
Hip Hop and Breakdance Instructor

- Taught weekly classes for groups of kids | Boys Hip-Hop ages 8-12 | Teen Hip-Hop

PROJECTS AND LEADERSHIP:

AI Dungeon Master | Full Stack Developer 2023

- **Created an AWS API to manage a real-time database abstracting OpenAI API calls and data management**
- **Used data science techniques to improve the memory of AI models by approximately 950%**
- **Used c# with Unity to create a 3D environment to immerse the user in the software**

Animatronics | Team Lead/Embedded Software and Electrical Lead **2021**

- Designed and manufactured FPGA circuits using Altium Designer to control 11 actuators simultaneously
- Used c++, and Node to write control software | utilized Wi-Fi and Serial Communication Protocall's

Rocket Avionics | Embedded Software and Electrical Lead 2020

- **Designed, manufactured, and programmed FPGA circuits to read extensive atmospheric data throughout the flight of a rocket and capture footage of the ground to identify targets mid-flight**

Personal Projects

- **Embedded Software Projects | Neopixel Lightsaber, Raspberry Pi Facial Recognition, Bluetooth Smart Home System, Reactive LED Sign** **2016-2023**
- **HackCU | Used c# with Unity to create a platformer game with AI controlled enemies in 24 Hours** **2022**
- **Web Development | Used ReactJS, HTML, and CSS to create a personal website (linked at top)** **2022**

School-Related Projects

- **Blockchain | Used different data structures in c++ to create a secure blockchain simulation** **2022**
- **Command Line Shell | Used c to create a command line shell with additional kernel control commands** **2023**
- **Travel Website| Used HTML, CSS, and Javascript to create a website to connect travelers with locals** **2022**

SKILLS AND INTERESTS:

Programming Languages: c++, Python, Java, c#, c, Javascript, React JS, CSS, HTML, SQL, Ruby, Scala
Computer Software: SOLIDWORKS, Adobe Illustrator, Adobe Photoshop, Docker, Firebase, GitHub, GitLab, AWS, Command Line Control, Linux development, Zephyr RTOS, CubeMX, ESP IDF, Free RTOS, MeshMixer, Cura, MarkForged, Excel, Google Applications
Proficient Operator of many Machines and Tools: Table/Miter Saw, Lathe, 3D printers, Circuit Boards (Nucleo, ESP32, Arduino, Raspberry Pi, etc.), Laser Engravers/Cutters, Milling Machines, Vinyl Printers
Personal Interests: Woodworking, Skiing, Dancing, Playing Guitar, Golfing, Maker Projects, Hiking, Board Games, Hockey